Kalyana Sundaram V

Game & Tools Engineer

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PROFESSIONAL SUMMARY

Game and tools engineer with experience leading rapid prototypes and building gameplay systems, Unity production pipelines, and native Windows, Android, and web products. Works across Unity/C#, Cocos/TypeScript, .NET, and React/TypeScript. Runs a multi-agent AI development workflow across Claude Code, OpenAI Codex, and GitHub Copilot with explicit file ownership, ticketed handoffs, and verification gates. Focused on readable architecture, testable automation, and fast delivery.

CORE SKILLS

Engines and languages: Unity 6 (URP), C#, Cocos Creator, TypeScript, JavaScript, Java, PowerShell

Gameplay and systems: gameplay programming, procedural generation, destruction and physics, deterministic simulation, economy and progression, UI navigation, Addressables

Tools and pipelines: Unity editor tooling, asset pipelines, Android build and release automation, content validation, automated testing (EditMode, PlayMode, Vitest)

Desktop, mobile, and web: .NET 8, WinUI 3, SQLite, native Android, React, Vite, Three.js, Tailwind CSS

AI-assisted engineering: Claude Code, OpenAI Codex, GitHub Copilot, OpenAI API, Gemini, ComfyUI; multi-agent orchestration, handoff protocols, structured AI code review

Services and collaboration: Firebase, Nakama, UniTask, Git, GitHub Actions, GitLab, Bitbucket, Jira, Confluence, Agile, technical leadership

PROFESSIONAL EXPERIENCE

Independent Game & Tools Engineer | Independent | March 2026-Present

- Designs and builds gameplay prototypes, Unity content pipelines, native Windows products, local-first Android tools, and a React/TypeScript portfolio; owns product definition, architecture, implementation, testing, and packaging.
- Runs parallel AI coding agents (Claude Code, OpenAI Codex, GitHub Copilot) with exclusive file ownership, isolated git worktrees, shared AGENTS.md entry points, and durable project memory so agents never edit the same file at once.
- Coordinates multi-agent delivery through HANDOFF.md boards that record goal, owner, status, blockers, exact commands, and resume steps; agents write a handoff before usage limits so another agent can continue without lost context.
- Connects agents to tickets through the ProjectPulse agent bridge: agents begin, review, or block tickets, queue commands while the app is offline, and return summaries, tests, and changed files; only human review closes work.
- Gates every agent change with evidence: Unity batch builds and EditMode/PlayMode results, standalone compile checks, headless generator harnesses, and lint, typecheck, and test steps in CI before commit or deploy.

Gameplay Engineer | Qila Games | November 2024-March 2026

- Contributed gameplay and product systems for Ludo Hero in a four-developer team working with artists.
- Worked across local and network board play, turn state, kingdom progression, rewards, mini-games, UI, asynchronous loading, analytics, authentication, and addressable live content.
- Used Unity, C#, UniTask, Nakama, Firebase, and Addressables; public descriptions intentionally omit private implementation details.

Prototype Team Tech Lead / SDE2 | Bombay Play | September 2021-May 2023

- Led engineering delivery across 12 rapid game prototypes in collaboration with game design and art.
- Built Word Wall and seven Match10 variations that advanced into production development, including content-generation, level, pacing, controls, effects, and power-up systems.
- Contributed to Bananagrams, Game of Life, Cube Match, Tower Clash, and Card Wars; supported prototype-to-production handoff, debugging, generated content, 3D math, session logic, and automated testing exploration.
- Coordinated work through Agile practices with Bitbucket, Jira, and Confluence.

Founder / Independent Developer | Ameya Game Studio (closed, part-time) | 2016-2021

- Founded and operated a small independent game studio alongside other work, owning game design, engineering, testing, release preparation, and product iteration.
- Published King Cube for Android and built additional mobile games and a marketing application prototype for small-business use.

Game Developer | URG Infotech | March 2018-April 2019

- Independently developed Android game prototypes and recreations of casino and popular-game mechanics.
- Worked across gameplay implementation, UI, debugging, device builds, and iterative delivery.

SELECTED PROJECTS

Ludo Demolition | Unity 6, URP, C#, Android

- Mobile-first 3D demolition game spanning contracts, tool and dice phases, destructible building parts, score, rewards, economy, progression, and road-aware traffic and ambient life.
- Deterministic world-generation package with canonical plan data, hashing, golden signatures, and an Android benchmark harness; organic settlements with rivers, bridges, quays, outskirts, parcel-safe footprints, and road-access checks.
- Authored-prefab destruction clusters driven by JSON settings, a production Android APK build pipeline, and mobile lighting tuned with Editor Device Simulator QA.
- About 40 Unity editor tools: AI style-guide and UI pipeline (OpenAI, Gemini, ComfyUI with approval gates), UI and popup batch builders, tool-effect and fracture authoring, settlement planning lab, Android RC metadata builds, 19-package systems exporter, validators, and Play Mode debuggers.

ProjectPulse | WinUI 3, .NET 8, C#, SQLite, Git, OpenAI API

- Native Windows engineering workspace for projects, tickets, Git commits and patches, file history, activity timelines, schedules, and insights with local-first SQLite persistence.
- Agent bridge and CLI let Claude Code, Codex, and Copilot register work against tickets and hand back reviewable evidence; OpenAI-backed slice review turns selected changes into structured implementation feedback.

AssetForge / PackTriage | Unity, C#, PowerShell

- Processes a catalog of 427 Unity asset packs through download intake, offline package scanning, Unity deep probes, verdict assignment, and auditable reports.
- Normalizes approved assets with dependency containment, GUID preservation and rewrites, URP material conversion, export manifests, and SHA-256 integrity verification.

Portfolio Site | React, TypeScript, Three.js, Vite, Tailwind CSS, GitHub Actions

- Interactive portfolio with project case studies, a tool catalog, and hosted resume at kalyanasundaramv.github.io; format, lint, typecheck, content and UI tests, and public-asset checks gate every GitHub Pages deploy.

Money Planner | Native Android, Java 17

- Local-first Android MVP that parses bank and card SMS on-device into spending, EMI, debt, savings, runway, and upcoming-obligation views, with parser confidence and no internet permission.

Wallpaper Workbench | WinUI 3, .NET, C#

- Windows image and video wallpaper product with configurable providers, resolution filtering, downloads, collections, rotation, fit modes, and native desktop playback.

Civitai Asset Manager | WinUI 3, .NET, C#, SQLite

- Local model-library manager for scanning, hashing, de-duplication, metadata enrichment, tags, trigger words, prompt recipes, downloads, and reversible cleanup.

SHIPPED AND TEAM GAMES

King Cube | Unity, C#, Android | Published

- Mobile action-puzzle game with grid movement, split-and-rejoin mechanics, laser objectives, generated level data, transforming floors, pooled hazards, and guided onboarding.

Ludo Hero | Unity, C#, Nakama, Firebase, Addressables | Team project

- Mobile board game with local and network play, turn-state systems, kingdom progression, rewards, mini-games, analytics, authentication, and live-content support.

Bombay Play prototypes | Word Wall | Match10 (seven variations) | Bananagrams | Game of Life | Cube Match | Tower Clash | Card Wars

EDUCATION

Bachelor of Engineering in Computer Science | Meenakshi Ramaswamy Engineering College | Affiliated with Anna University

LANGUAGES

English | Malayalam | Tamil | Hindi